ACROPOLIS INSTITUTE OF TECHNOLOGY AND RESEARCH

Department of Information Technology

Synopsis

on

Intelligent Real-Time Street Parking Solutions-ParkHub

1. INTRODUCTION

1.1. Overview

As cities grow in population and the number of vehicles in India spikes, parking space has turned into a serious challenge for authorities and citizens, making the challenge multifaceted. On-street parking space is limited, and the continuous search for free parking spots creates traffic and environmental issues. The problem of user frustration gets compounded. The Smart Vehicle Parking App solves this problem and enables parking lot administrators to monitor open parking spaces for users to reserve and access the parking facilities. The users have access to a simple interface, making the process seamless. Parking swapping reduces the time spent parking and enhances space utilization while in a span of minutes the entire process is automated, self-optimizing the cost and revenue. This app has the potential to bring transparency, and the goal is to bring the much-needed technology for better organization in the everyday city parking systems.

1.2. Objective

The primary goal of this project is to create a smart and easy-to-use parking management system that benefits both administrators and citizens. For administrators, the objective is to manage parking lots efficiently, monitor occupancy, and earn revenue through optimal utilization of available space. For users, the objective is to offer an easy method to reserve, occupy, and vacate parking spaces without inconvenience. By minimizing search time, managing congestion, and providing fair pricing, the project aims to make city parking more structured, transparent, and sustainable.

2. LITERATURE SURVEY

2.1. Existing Problem

Parking infrastructure and technology already exist across Indian cities, though most lack the simplicity and accessibility central to our proposed web-based system. Below are comparative case studies of five existing solutions, along with their merits, demerits, and identified gaps.

1. **NDMC's Smart Parking Management (New Delhi)**
   - **Problems addressed:** Lack of real-time occupancy data, fragmented management of 155 parking facilities.
   - **Advantages:** Provides vehicle flow tracking, slot availability, revenue monitoring, and online payments.
   - **Disadvantages:** Previous efforts delayed; high complexity, reliant on a dedicated agency.
   - **Gaps identified:** No user-facing reservation system or dynamic pricing; lacks a simple web interface.
2. **Fully Automated Multi-Level Parking (Hyderabad,**
   - **Problems addressed:** Severe congestion in dense urban parking zones.
   - **Advantages:** Zero human intervention , sensor-based slot allocation , with German Palish tech. Fast and accessible
   - **Disadvantages:** Very high initial cost (₹102 crore);limited to one location
   - **Gaps identified:** No web-based booking system, heavily reliant on costly physical infrastructure.
3. **Puzzle Parking (Noida Sector 63)**
   - **Problems addressed**: High demand and limited space in office districts.
   - **Advantages**: Space-efficient design, automated retrieval in 3–6 minutes, enhanced safety.
   - **Disadvantages**: High installation cost (₹16.5 crore); limited to dense office hubs.
   - **Gaps identified**: No dynamic pricing or web-based reservation system; hardware intensive.

4. **Hydraulic Multi-Level Parking (Patna, Maurya Lok)**
    - **Problems addressed**: Chaotic roadside parking and shortage of structured parking.
    - **Advantages**: Mechanized shuttle system; supports pre-booking and enforcement.
    - **Disadvantages**: Poor adoption despite ₹28 crore investment; under-utilized.
    - **Gaps identified**: Absence of a user-friendly booking dashboard to drive usage.

2.2 Project Description

The working of the project follows a systematic flow of operations between the Admin and User, ensuring smooth management of parking lots and spots. The following flowchart illustrates the step-by-step process of how the application functions, from login to parking spot allocation and release.

Features:

- An **Admin** account is created by default at the start of the app if no admin exists. Admins cannot register themselves through the UI.
- **User Registration & Login:** Users can register with name, email, and password. Users can log in to access the dashboard.
- **Parking Spot Booking:** Logged-in users can view available parking lots, book an available spot, and release it when done. The system calculates total cost based on hours parked.
- **Admin Dashboard:** Admin can create, edit, and delete parking lots. The Parking Lot can delete only if all spots in the parking lot are empty. For each lot, they can manage the number of spots and view detailed information.
- **Booking History:** Users can view their full booking history along with duration and charges.
- **Charts and Analytics:** Admins can view bar charts of available and occupied spots across lots. Users get insights like most visited locations, total hours parked, and amount spent.
- **Search Features:** Admins can search for parking spots by lot name or spot ID and view the current booking user if occupied.

- **Security:** Session-based access control is implemented to restrict views to respective roles (admin or user).

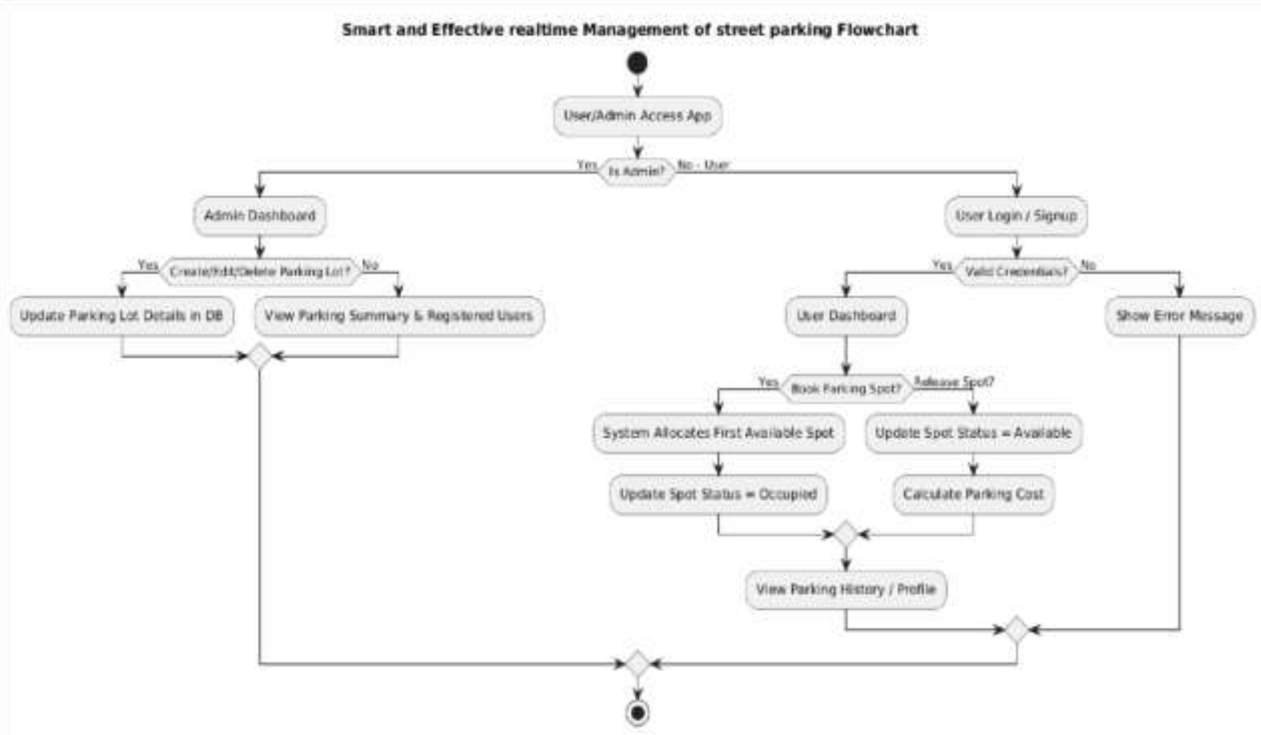

Fig1 : Flowchart of Smart and Effective Real time Street Parking System

3.THEORETICAL ANALYSIS

3.1. Hardware/Software Designing

Hardware Requirements: The solution will require a robust server infrastructure to handle data storage, processing, and the computational demands of the machine learning model. This server should have enough RAM, a high-performance CPU, and scalable storage. Client-side hardware will be minimal, as the application is web-based and can be accessed from any standard computer, laptop, or mobile device with a modern browser.

Software Requirements:

Category	Requirement	Description / Purpose
Frontend	React.js	For building dynamic, responsive parking dashboards, booking forms, and admin panels.
	HTML5, CSS3, JS	Core web technologies for structure, styling, and interactivity.
Frontend	Chart.js (via react-chartjs-2)	To visualize parking spot occupancy, usage trends, and revenue with interactive charts.
Backend	Node.js + Express	To handle server-side logic, APIs, and manage requests.
Database	MongoDB	NoSQL database to store parking spots, bookings, users, and transaction records.
ORM/ODM	Mongoose	Schema definition, validation, and interaction with MongoDB collections.
Development Tools	VS Code	Code editor for development.
	GitHub/Git	Version control and collaboration.
	Postman	API testing and debugging.
Hosting/Deployment	Heroku / Render / AWS	For hosting the web application online.
Browser Support	Chrome, Edge, Firefox	Ensures compatibility across major browsers.

4.APPLICATIONS

1. Municipal Street Parking Management

Provides city municipal corporations with a centralized system to manage on-street parking zones, monitor occupancy, and generate revenue reports in real time..

2. Educational Institution Campus Parking

Helps colleges and universities manage separate parking allocations for faculty, students, and visitors, reducing congestion during peak hours.

3. Commercial Complex and Mall Parking

Enables shopping malls and commercial buildings to manage paid customer parking with automated hourly billing and quick spot lookup.

4. Office Park and IT Hub Parking Management

Allows large office complexes to allocate and monitor parking for employees and visitors, with analytics to support future infrastructure planning.

5. Residential Society Visitor Parking

Offers gated communities a simple way to track and manage visitor parking, maintaining clear records of vehicle entry, exit, and duration.

6. Foundation for Smart City Integration

Acts as a scalable digital parking module that can be extended with IoT sensors and real-time tracking as part of future smart city initiatives.

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